

# Kinetic Theory of Gases

## 3. Kinetic Interpretation of Temperature

According to assumption of Kinetic theory of gases, there is only translation motion of the molecule and there is not any potential acting between them, so

Average energy  $\langle E \rangle$  of gases are equivalent to Average translation energy of a molecule

$$\langle E \rangle = \frac{1}{2} m \langle v^2 \rangle$$

$$\text{Pressure at } P \text{ as } P = \frac{1}{3} m n \langle v^2 \rangle = \frac{2}{3} n \left( \frac{1}{2} m \langle v^2 \rangle \right) = \frac{2}{3} n \langle E \rangle$$

$$PV = \frac{2}{3} V n \langle E \rangle \quad PV = \frac{2}{3} N \langle E \rangle \quad \text{where } n = \frac{N}{V} \text{ number density}$$

$$\langle E \rangle = \frac{3}{2} \frac{RT}{N_A} \text{ and } \langle E \rangle = \frac{3}{2} \left( \frac{R}{N_A} \right) T$$

$$\langle E \rangle = \frac{3}{2} K_B T \quad \text{where } k_B \text{ is Boltzman constant}$$

So average kinetic energy is given by

$$\langle E \rangle = \frac{3}{2} k_B T \quad \text{where T is absolute temperature.}$$

**Example:** It is possible to treat electromagnetic radiation in container whose wall is mirrors, as a gas of particle (photons) with a constant speed  $c$  and whose energy is related to their momentum  $p$  which is directed parallel to their velocity by  $E = pc$ . Show that if container is

full of radiation the equation of state is  $PV = \frac{1}{3} E$

**Solution:** Pressure  $P = \frac{1}{3} nm \langle v^2 \rangle = \frac{1}{3} n \langle mv \cdot v \rangle = \frac{1}{3} n \langle \vec{p} \cdot \vec{v} \rangle$

For Photon  $v = c$  and velocity is parallel to momentum, so

$$P = \frac{1}{3} n \langle Pc \rangle \quad \Rightarrow \quad P = \frac{1}{3} \frac{N}{V} \langle pc \rangle$$

$$PV = \frac{1}{3} \langle Npc \rangle \Rightarrow \quad PV = \frac{1}{3} E$$